

Md. Almabud **JUWAL**

Senior Software Engineer | Backend, Data & Architecture

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 Dhaka, Bangladesh

 SUMMARY

Senior Software Engineer with more than six years of experience building backend and data-intensive systems in Python, FastAPI, Django, and PostgreSQL. Experienced in system architecture, performance optimization, asynchronous data pipelines, and leading small engineering teams. Recent work includes a machine-learning pipeline for entity matching and an MCP server for language-model report generation.

 SKILLS

Programming	Python, SQL, JavaScript/TypeScript, Go, C/C++
Backend & Frameworks	FastAPI, Django, Django REST Framework, Celery
Data & Databases	PostgreSQL, Pandas, Redis, MongoDB, Neo4j, Elasticsearch
Machine Learning & AI	scikit-learn, Sentence Transformers, RapidFuzz, Azure Document Intelligence, MCP
Infrastructure & Engineering	Docker, RabbitMQ, Git, CI/CD, Jenkins, Ansible
Frontend	Angular

 EXPERIENCE

Present Jun 2024	Senior Software Engineer Cogwheel Analytics, Colorado, United States (Remote) - Architect the next-generation Cogwheel Analytics platform as a modular monolith following Clean Architecture principles, with reusable handling for authentication, pagination, and request validation that reduces repetitive implementation work - Lead the development and maintenance of the Data Cleaning service, implementing data-validation and transformation workflows to improve data integrity and consistency - Designed and implemented a hotel-name matching pipeline that combines TF-IDF and fuzzy-string retrieval, sentence-embedding search, and cross-encoder re-ranking with threshold-based acceptance - Built an MCP server that lets a language model generate custom reports from analytics data, and worked on extracting data from PDF reports with Azure Document Intelligence - Design and implement backend services, APIs, and asynchronous data-processing and web-scraping workflows Python FastAPI PostgreSQL Pandas Celery Scrapy scikit-learn Sentence Transformers MCP
May 2024 Jul 2023	Senior Software Engineer Brain Station 23 Ltd, Dhaka, Bangladesh - Led the turnaround of a delayed project by mentoring 3 junior engineers, restructuring task allocation, and redesigning the system architecture, enabling successful delivery of the second version within 6 months - Designed software architectures for Task Station and Sayvery, with focus on scalability, maintainability, and clear separation of responsibilities - Managed the technical backlog by translating product and stakeholder requirements into prioritized engineering tasks aligned with project goals and delivery timelines - Designed and implemented web-scraping and invoice-parsing solutions to automate data collection and extraction Python Django Django REST Framework PostgreSQL Celery Scrapy
Jun 2023 Oct 2021	Software Engineer Brain Station 23 Ltd, Dhaka, Bangladesh - Improved system performance by approximately 10% through database-query optimization and API response time by approximately 5% through application-level caching - Upgraded Python and Django versions and refactored the existing codebase - Contributed to real-time synchronization between backend and frontend systems using WebSockets and Go, while mentoring an Associate Software Engineer Python Django Django REST Framework PostgreSQL Redis Celery Go

Sep 2021 Jul 2020	Junior Software Engineer Tirzok Private Limited, Dhaka, Bangladesh ➤ Reduced processing time for 3.3 million records from approximately 29–35 minutes to 2–4 minutes using PostgreSQL COPY and database triggers ➤ Reduced database query response time by approximately 50% through query optimization ➤ Contributed to backend and database architecture, including the implementation of a custom authentication backend and graph-based data solutions using Neo4j ➤ Automated deployment workflows using Jenkins and Ansible Python Django PostgreSQL Redis Neo4j Jenkins Ansible Angular
Apr 2020 Dec 2019	Web Developer Wizdoor Inc, Dhaka, Bangladesh ➤ Contributed to backend and database architecture while coordinating implementation with frontend and design teams ➤ Developed a real-time chat application using Django Channels and WebSockets Python Django Django REST Framework PostgreSQL Django Channels Angular

PROJECTS

REHIVE

Oct 2021 – Feb 2023

<https://rehive.com/>

Contributed to a South Africa-based fintech platform supporting multiple currencies, including cryptocurrency. Implemented the billing service and contributed to a plugin-based architecture that enables additional services and commerce-oriented functionality to be integrated with the platform.

Python Django Django REST Framework PostgreSQL Celery RabbitMQ Redis Docker Angular

LEGISLIB

May 2020 – Jul 2020

Legal-document processing and search platform for case-law materials from Bangladesh, India, the UK, and the USA. Processed and reformatted PDF documents and indexed legal content to provide searchable access for Bangladeshi legal professionals and students.

Python Django Django REST Framework MongoDB Elasticsearch PDF Processing

EDUCATION

2020 2016	B.Sc. in Computer Science and Engineering ➤ Daffodil International University, Dhaka, Bangladesh ➤ CGPA : 3.77 / 4.00 ➤ Final-year project (A+) : mobile application that reads exam marks from photographed cover pages with the Microsoft Azure Vision API
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CERTIFICATIONS & COURSES

Jun 2025	Advanced Certificate for Management Professionals (ACMP 4.0), University of Dhaka
Apr 2023	AWS Certified Cloud Practitioner
2021	Object-Oriented Design, Coursera
2021	The Ultimate Docker Course, Mosh Hamedani

ACADEMIC & PROGRAMMING ACHIEVEMENTS

2018	Participant, ACM ICPC Dhaka Preliminary Contest
2018	4th place , DIU Intra Team Programming Contest
2017	2nd place , DIU Intra Team Programming Contest
2017	9th place , DIU Takeoff Individual Programming Contest

LANGUAGES

Bengali : Native **English** : IELTS Academic 6.5 (CEFR B2) **Hindi** : Conversational